

Performance Testing

Date	15 March 2026
Team ID	SWTID-2026-9325
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	User Login, AI Debt Analysis API, Recommendation API
Test Environment	Local System (Flask Backend + React Frontend)
Test Date	10 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login	50	60	All users login successfully with low response time
2	AI Debt Analysis	30	90	AI generates debt analysis without timeout
3	Generate Debt Relief Recommendations	25	60	Personalized recommendations generated successfully
4	Multiple Users Access DashBoards	100	200	Dashboard remains responsive without server crash

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.18 sec	Pass	Fast Response
2	Response Time (Max)	< 5 seconds	3.42 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)	20 req/sec	27 Req/sec	Pass	Good Throughput
4	Error Rate	< 1%	0%	Pass	No request failures
5	CPU Utilization	< 80%	56%	Pass	Stable CPU usage
6	Memory Utilization	< 80%	63%	Pass	Memory usage under limit

Step 4: Observations & Analysis

- The AI-powered Debt Relief Platform performed efficiently under normal and moderate load conditions.
- User authentication and debt analysis modules responded within the expected time limits.
- No request failures or server crashes occurred during testing.
- CPU and memory utilization remained below the defined thresholds.
- The system successfully handled concurrent users while maintaining stable performance.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

